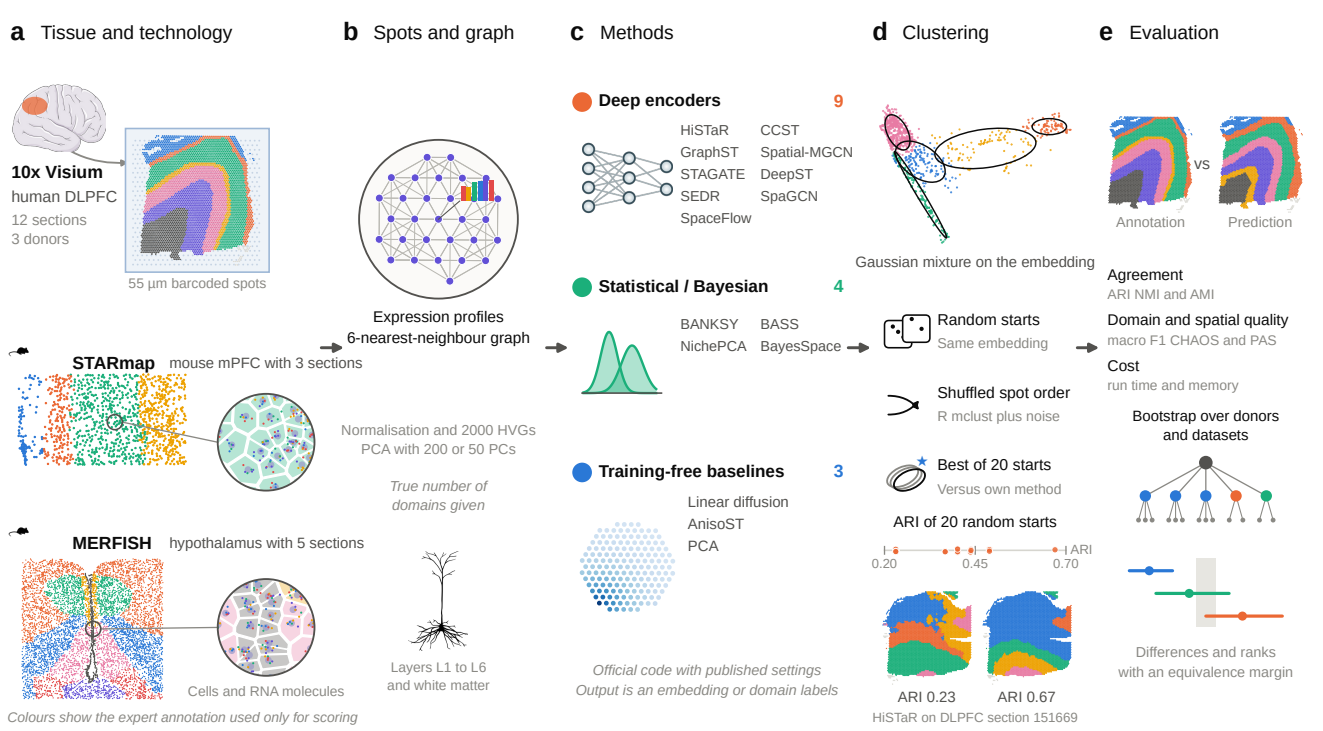

a Tissue and technology
10x Visium
human DLPFC
12 sections
3 donors
55 µm barcoded spots
STARmap mouse mPFC with 3 sections
MERFISH hypothalamus with 5 sections
Cells and RNA molecules
Colours show the expert annotation used only for scoring
b Spots and graph
Expression profiles
6-nearest-neighbour graph
Normalisation and 2000 HVGs
PCA with 200 or 50 PCs
True number of domains given
Layers L1 to L6 and white matter
c Methods
Deep encoders 9
HiSTaR
GraphST
STAGATE
SEDR
SpaceFlow
CCST
Spatial-MGCN
DeepST
SpaGCN
Statistical / Bayesian 4
BANKSY
NichePCA
BASS
BayesSpace
Training-free baselines 3
Linear diffusion
AnisoST
PCA
Official code with published settings
Output is an embedding or domain labels
d Clustering
Gaussian mixture on the embedding
Random starts
Same embedding
Shuffled spot order
R mclust plus noise
Best of 20 starts
Versus own method
ARI of 20 random starts
ARI
0.20
0.45
0.70
ARI 0.23
ARI 0.67
HiSTaR on DLPFC section 151669
e Evaluation
vs
Annotation
Prediction
Agreement
ARI NMI and AMI
Domain and spatial quality
macro F1 CHAOS and PAS
Cost
run time and memory
Bootstrap over donors and datasets
Differences and ranks with an equivalence margin